DR. MUBASHIR H. WANI, PHD, CENG, PMP

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Electrical Engineer | Computational Intelligence for Energy Systems

Profile

I am a Chartered Electrical Engineer (CEng) and Automation Engineer with a PhD in Electrical and Computer Engineering from the University of Auckland. My work integrates engineering practice, computational intelligence, and energy system optimization, with a focus on intelligent building systems, digital twins, and sustainable infrastructure.

Currently working with Jacobs Engineering in Dubai, I contribute to large-scale international infrastructure projects including NEOM developments, where my work involves electrical power distribution, fire life safety systems, ELV/ICT infrastructure, and multidisciplinary system integration. This industry experience informs my academic research and teaching by connecting theoretical concepts with real-world engineering challenges.

My research focuses on the application of computational intelligence and control methodologies to building energy management systems (BEMS), smart buildings, and cyber-physical infrastructure systems. During my PhD, funded by the Ministry of Business, Innovation and Employment (MBIE), New Zealand, I developed advanced optimization and surrogate-modeling frameworks for energy-efficient building operation and intelligent infrastructure design.

In addition to research, I am passionate about teaching electrical engineering, energy systems, control systems, and intelligent infrastructure technologies, and about mentoring students in applying analytical and computational tools to solve real engineering problems.

My professional and academic interests lie at the intersection of electrical power systems, intelligent control, sustainable buildings, and computational intelligence for infrastructure systems.

Core Competencies

- Computational Intelligence (CI) and Intelligent Algorithms (IA)
- Electrical System Design (ELV/ICT, Load Analysis)
- APC & Automation (PID, MPC, Supervisory Logic)
- Energy Simulation (EnergyPlus, MATLAB, Python)
- Demand-Side Management (DSM) & Demand Response (DR)
- Digital Twins & Co-Simulation Frameworks
- Optimization Algorithms (NSGA-II, AI Surrogates)
- Renewable Integration (PV, Wind, BESS - conceptual)
- Automation & BMS/BEMS Integration

Research Interests

- Smart Grid and Energy Optimization
- Surrogate and First-Principles Mathematical Modeling of Dynamic Systems, such as Buildings, HVAC Systems, and Other Energy Systems
- Multi-Objective Optimization for Building Energy Systems
- AI and Machine Learning for Sustainable Energy Solutions
- Formal Modeling and Simulation (EnergyPlus, MATLAB, CFD)
- Control Strategies for HVAC and Renewable Energy Integration
  Digital Twins and Cyber-Physical Systems for Smart Cities

Distinctions/Honours

1. **Faculty of Engineering and Design Elite MSc Scholarship** - University of Bath, UK (2015–2016) **[Sponsor License: C2DENCHX0]**
2. Receiver of ***Callaghan Innovation R&D Fellowship Grant*** (2019-2022) from the Ministry of Business, Innovation and Employment (MBIE), New Zealand.
3. **Outstanding Paper Award** (Titled: "***Accelerating Building Energy Simulations with PyE+: A Surrogate-Driven AI Framework for Real-Time Decision-Making***") – IEEE PEET 2025, Shiga, Japan.
4. **Best Oral Presentation Award** (Titled: "***Accelerating Building Energy Simulations with PyE+: A Surrogate-Driven AI Framework for Real-Time Decision-Making***") – IEEE PEET 2025, Shiga, Japan.

Research Experience

Research & Development Fellow – Building Energy Management Systems (BEMS)

Fisher & Paykel Technologies Limited, Auckland, New Zealand (2020-2022)

- Developed a **multi-objective optimization framework** for smart energy management in buildings.
- Designed AI-driven control strategies balancing **energy consumption** and **thermal comfort**.
- Integrated **formal modeling techniques** to improve predictive accuracy in energy simulations.
- Published findings in high Impact Factor (IF) journals like **Annals of Operations Research**, **Electronics**, and **Building and Environment.**

Graduate Research Assistant

Department of Electrical, Computer, and Software Engineering, The University of Auckland (2018-2022)

- Conducted research on **meta-heuristic algorithms for thermal parameter estimation**.
- Developed **co-simulation models (MATLAB & EnergyPlus)** for optimizing HVAC systems.
- Published multiple IEEE and Springer research papers on **energy-efficient building control**.

Publications and Research Output

Peer-Reviewed Journal Articles

1. **Wani, M., Hafiz, F., Swain, A., & Broekaert, J.** (2023). *Balancing Energy Consumption and Thermal Comfort in Buildings: A Multi-Criteria Framework*. **Annals of Operations Research**. DOI: https://link.springer.com/article/10.1007/s10479-023-05747-y
2. **Wani, M., Hafiz, F., Swain, A., & Ukil, A.** (2023). *A Multi-Objective Approach to Robust Control of Air Handling Units for Optimized Energy Performance*. **Electronics**. DOI: https://www.mdpi.com/2079-9292/12/3/661
3. **Wani, M., Swain, A., Ukil, A., Ploder, M., & Koole, R.** (2022). *Optimizing the Performance of Forced Extraction Systems: A Multi-Objective Framework*. **Building and Environment**. DOI: https://www.sciencedirect.com/science/article/pii/S0360132322004504
4. **Wani, M., Hafiz, F., Swain, A., & Ukil, A.** (2019). *Parameter Estimation of Thermal Models using a Meta-Heuristic Approach*. **Energy and Buildings**. DOI: https://www.sciencedirect.com/science/article/pii/S0378778820316832

Conference Proceedings

1. **Wani, M., Hafiz, F., & Swain, A.** (2025). *Accelerating Building Energy Simulations with PyE+: A Surrogate-Driven AI Framework for Real-Time Decision-Making*. **PEET 2025 - International Conference on Power Engineering and Electrical Technology - Japan.** DOI: https://ieeexplore.ieee.org/document/11340811
2. **Wani, M., Swain, A., & Ukil, A.** (2021). *Intelligent Controller for Thermal Comfort Management in Buildings*. **IECON 2021 - IEEE Industrial Electronics Society - Canada.** DOI: https://ieeexplore.ieee.org/abstract/document/9589666
3. **Wani, M., Swain, A., & Ukil, A.** (2019). Control Strategies for Energy Optimization of HVAC Systems in Small Office Buildings using EnergyPlus. **ISGT 2019 - Innovative Smart Grid Technologies - Asia.** DOI: https://ieeexplore.ieee.org/abstract/document/8880806

Ongoing/Accepted Research Outputs

1. **Monograph (Book):** Extended version of my PhD Thesis titled as: **"Digital Twins for Building Optimization: Enhancing Energy and Comfort"** to be published in *Springer-Nature's Studies in Infrastructure and Control* series (The book is in publication process, will be available by September, 2026)
2. **Journal/Article: "Towards Sustainable Decision-Making in Multi-Criteria Building Design: A Game-Theoretic Approach",** Omega, Elsevier [Accepted: 13th April, 2026]
3. **Conference Paper: "PyE+: An Event-Driven Hardware-in-the-Loop Digital Twin Framework for Building Energy Systems with MQTT-Based Signal Integration and Real-Time EnergyPlus™ Co-Simulation",** PEET Japan, IEEE [Accepted: 30th July, 2026]

Education

Nov 2018 - Oct 2022 — The University of Auckland, Auckland, New Zealand

Doctor of Philosophy (PhD) in Electrical & Electronic Engineering

- My research focused on Building Energy Management Systems (BEMS), with regard to Demand Side Management (DSM) and Demand Response (DR) - which are quite important from Smart Grids perspective. The research particularly involved optimizing building energy consumption, by resorting to modern and advanced control theory on:
  - Building HVAC Systems
  - Lighting Systems
  - Domestic and Commercial Appliances
- **Thesis Title:** "Intelligent Control Strategies for Efficient Building Energy Management Systems (BEMS)"
- Developed computational models to optimize energy efficiency while ensuring occupant comfort.
- Funded by Ministry of Business, Innovation and Employment (MBIE), New Zealand via Callaghan Innovation.
- **Accreditation:** New Zealand Qualifications Authority (NZQA)

Sep 2015 - Sep 2016 — University of Southampton, Southampton, United Kingdom

M.Sc in Energy and Sustainability with Electrical Power Engineering

- **Modules:** Power System Analysis, Power Generation-Technology & Impact on Society, Power Transmission & Distribution, Fundamental Principles of Energy, Advanced Electrical Materials, High Voltage Insulation Systems and Power Electronics for DC transmission.
- Specialization in Renewable Energy Systems and Power Networks.
- **M.Sc. Project:** (Jun 2016 - Sept 2016) "Space Charge Behaviour in Multi-Layered Dielectrics".
- **Grade:** 76.5% (Distinction - Achieved)
- **Accreditation:** Accredited by IET; Section: CEng, Course Reference: 12950

Sep 2010 - May 2015 — University of Kashmir, Srinagar, India

B.Eng in Electronics & Communication Engineering

- **Core modules included:** Electronics (I-II), Electrical Measurements, Electronic Measurements & Instrumentation, C-programming, FORTRAN, Communication Systems (I-II), Radar Systems, Microprocessors, Digital Electronics, Data Communications, Optical Fiber Communication, Information Theory & Noise, Power Electronics, Power Systems (I-II), Electrical Machines (I-II), Network Analysis, Control Systems, Principles of Electrical Engineering, Mathematics (I-V), Physics.
- **Dissertation:** (Dec 2014 - Mar 2015) "CPLD Based Smart Energy Distribution and Fault Detection System".
- **Grade:** 79% (Distinction - Achieved)
- **Accreditation:** Accredited 'A+' by NAAC, AICTE.

Professional Experience

Electrical/ELV Design Engineer

Jacobs Engineering Group Inc., Dubai, UAE (2023 – Present)

- Contributing to major international projects including **Islamic Civilization Village (ICV), NEOM Time Travel Tunnel (TTT), NEOM Sindalah Island, NEOM Vault, Al Jalila Hospital, HUMAIN Data Center, SB4 and Qiddiya Investment Company (QIC)** developments.
- Responsible for the design and coordination of Electrical, ICT, ELV, Security, Fire Alarm (FDAS), PAVA, and Building Management System (BMS) solutions across multidisciplinary projects.
- Perform electrical load calculations, equipment sizing, space planning, and system integration studies in accordance with project and authority requirements.
- Lead QA/QC reviews, BIM coordination, clash-resolution workshops, and interdisciplinary design coordination to ensure technical compliance and optimized project delivery.
- Develop Fire Alarm (FDAS) and PAVA system architectures, including integration and interfacing with BMS and other life-safety systems.
- Prepare Detailed Design (DD) documentation, including Cause & Effect Matrices, Basis of Design (BoD) Reports, Detailed Design Reports, and technical specifications.
- Author project specifications (CSI format), including **Section 25 10 00** – Integrated Automation Network Equipment and **Section 28 46 21.11** – Fire Detection and Alarm Systems (FDAS).
- Design structured ICT and ELV cabling infrastructure, including copper (Cat6A/Cat7) and fiber-optic (OS2/OM5) networks, containment routing strategies, cable sizing, tray sizing, and termination methodologies.
- Generate Single-Line Diagrams (SLDs) for ICT, Security, Fire Alarm, and PAVA systems at both building and campus/site-wide levels.
- Perform Sound Pressure Level (SPL) modeling and assessments for PAVA systems to verify intelligibility and code compliance.
- Conduct Wi-Fi network planning and optimization using Ekahau, including heat-map analysis, coverage validation, and performance assessment.
- Perform battery autonomy calculations, Voltage-Drop (VD) analysis, UPS sizing, and backup power assessments for life-safety and critical systems.
- Coordinate closely with architectural, interior design, BIM, Fire & Life Safety (FLS), client, authority, and supplier teams throughout all project phases.

Solutions Engineer- Electrical Power Management Systems (EPMS)

Electrotest Limited, Auckland, New Zealand (2022 – 2023)

- Delivered technical solutions for power management systems, improving client satisfaction and operational efficiency.
- Performed **UPS sizing and battery bank calculations** for critical power applications.
- Managed **installation and commissioning** of single-phase and three-phase UPS systems (ranging from 1kVA to 60kVA modular UPSs).
- Led the **scoping of UPS equipment** for both new installations and maintenance engagements.
- Supervised **preventive and corrective maintenance** activities.
- Developed customized **maintenance and service agreements** for key clients.
- Conducted face-to-face technical meetings with major clients, including **Air New Zealand**, District Health Boards, and network engineering teams.

Graduate Teaching Assistant

The University of Auckland, Auckland, New Zealand (2018 - 2022)

I worked as a Graduate Teaching Assistant (GTA) in the School of Electrical Computer and Software Engineering (ECSE). Modules for which I provided support, included:

- COMPSYS723 (Embedded Systems Design)
- ELECTENG309 (Power Apparatus and Systems)
- COMPSYS704 (Advanced Embedded Systems)
- ELECTENG303 (Systems and Control)

Embedded Systems Lab Technician

The University of Auckland, Auckland, New Zealand (2019 – 2020)

- Designed experimental setups for **control systems & embedded electronics**.
- Revived FESTO Modular Production System (MPS) 500 using Beckhoff controllers and TwinCAT automation software for decentralized control.
- Provided guidance on **simulation-based learning** in MATLAB and Python.

Electrical Automation Engineer

Teknocrat's Control Systems (I) Pvt Ltd, Mumbai, India (2017 – 2018)

- Designed and implemented automation solutions from **concept through final delivery** for various industrial applications.
- Developed **PLC automation logic using Ladder Logic**.
- Gained hands-on experience with **HMIs** and **CNC-based plasma cutting machines**.

- Performed **system integration**, combining PLCs, HMIs, sensors, actuators, and control components into complete working automation systems tailored to client requirements.

Assistant Professor - School of Electrical and Electronic Engineering

SSM College of Engineering and Technology, J&K, India (2017)

- Taught courses in **electrical engineering: Communication Systems, Basics of Electrical Engineering, Analog Electronic Circuits, Solid State Physics.**
- **Supervised UG Capstone Projects.**

External Liaison Officer

IEEE Student Committee, University of Southampton, UK (2015 - 2016)

- Actively participated in organizing number of technical events.
- Responsibly liaised with the members of top organizations and invited them to provide technical talks to students.
- Successfully secured sponsorship from range of organizations, amounting to about £4000

Intern

Ritusha Consultants Private Ltd. (RCPL), Noida, India (2015)

- **Technical Support Executive:** Worked as a Technical Support Executive during the first phase of internship program in order to become familiar with the working environment.
- **Lead Trainer:** During second phase, as a lead trainer shared my understanding/knowledge of Embedded Systems and Robotics with final year engineering students - who were undergoing training, in this field - through proper lectures and assessed them by providing proper coursework, lab assignments in addition to conducting a computer based exam, for providing them a solid grounding in this field.
- **Outcome:** Developed strong working knowledge of basic and advanced embedded systems in addition to C-programming

Industrial Training and Certificate Courses

Apr 2025 - Jun 2025

Trainee

Arabian Infotech Training Institute L.L.C, Dubai, UAE

Received training on ETAP (Electrical Transient & Analyzer Program) Software. The 6 week traning course focused on, but was not limited to, the following:

- Practical training in AC/DC power system modeling and analysis using ETAP
- Load flow analysis, IEC 60909 short-circuit studies, and protective device coordination
- Transient stability analysis, harmonic analysis, grounding grid systems, and capacitor optimization
- Applied case studies for transmission, distribution, substations, and industrial power systems

Oct 2017 - Dec 2017

Trainee

Teknocrats Academy of Automation and Control Technology (TAACT), Mumbai, India

The training was delivered on two important sections of Electrical Automation Engineering:

- **Electrical Hardware Logic:** This involved training on control panel designing - using automation components like relays, timers, counters, contactors, momentary and permanent NO/NC contacts etc. Apart from that, programming of SINAMICS V20 AC Drives, to control three-phase induction motors for automation purposes was also performed, in addition to using protection devices - like TOLRs, MCBs, MCCBs, MPCBs etc. - for their safe operation.
- **Software Training:** Major portion of the training focused on programming of PLCs in Allen Bradley (Micrologix 1400) and Siemens (S7 300) platforms - for various automation projects, assigned to us during the training. Final segment of training included interfacing of PLCs with SCADA (Supervisory Control and Data Acquisition) and HMI (Human Machine Interface).
- **Outcome:** A solid grounding in electrical hardware logic design, significant exposure to PLC programming and troubleshooting was achieved, along with specific skills, necessary for project management.

Jan 2014 - Jan 2014

Trainee

Hewlett Packard (HP), Noida, India

- Gained understanding of various simple electronic gadgets/components like LED's, transistors, transceivers, PIR's, piezoelectric crystals, ADCs, LCDs etc, to name a few.
- Learnt programming of micro-controllers (ATmega 16 and 32 bit) and interfacing the above electronic gadgets/components to them in IDE (Integrated Development Environment), via C-programming.
- Obtained basic understanding of developing Embedded Systems for performing various dedicated functions, on demand.

Jan 2013 - Jan 2013

Trainee

Jammu & Kashmir Power Development Department (JKPDD), Srinagar, India

- Completed training at the 132/33 kV Grid Station, building practical field experience in power system operations and emergency response.
- Responded to electrical emergencies, including power cuts and electrical faults, while supporting continuity and safety of grid operations.

- Monitored feeder performance through control panels installed at the grid station to support operational oversight and system reliability.
- Developed hands-on understanding of the operation of electrical machines, including power transformers, SF6 circuit breakers, and current and potential transformers.
- Supported day-to-day grid station activities while strengthening technical knowledge of electrical distribution and substation equipment.
- Gained on-field exposure to fault handling and equipment monitoring within a live power development department environment.

Technical Skills

- **First-principles Modeling & Simulation:** EnergyPlus, MATLAB/Simulink, MLE+, Simcenter FloVENT (CFD), ETAP (Electrical Software)
- **AI & Machine Learning:** Python (Scikit-learn, TensorFlow), Meta-Heuristic Algorithms
- **Optimization Techniques:** Multi-Objective Optimization, Decision-Making Algorithms
- **Building Energy Systems:** HVAC Control, Smart Grid Integration, BMS & IoT
- **Programming & Tools:** Python, MATLAB, FORTRAN, Revizto (BIM), SketchUp, EnergyPlus, AutoCAD, Revit (basic understanding), DIALux, Bluebeam (Revu)
- **PLC Programming:** RS Logix 500, Simatic Manager (S7 300)
- WinCC Explorer for SCADA and WinCC Flexible for HMIs
- Able to fabricate simple and complex PCBs
- Control Panel designing and electrical hardware logic

Certifications and Memberships

- **Chartered Engineer (CEng)** – Institution of Engineering and Technology (IET), UK **[Registration Number: 715435**]
- **Project Management Professional (PMP)** – PMI, UAE Chapter **[PMP Number: 3697204]**
- **Member, IEEE Power and Energy Society (PES)**
- **Member of Society of Engineers (SoE), Dubai, UAE**

Academic References

- **Dr. Akshya Swain:** The University of Auckland; E-mail: a.swain@auckland.ac.nz; Tel: +642102340207
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